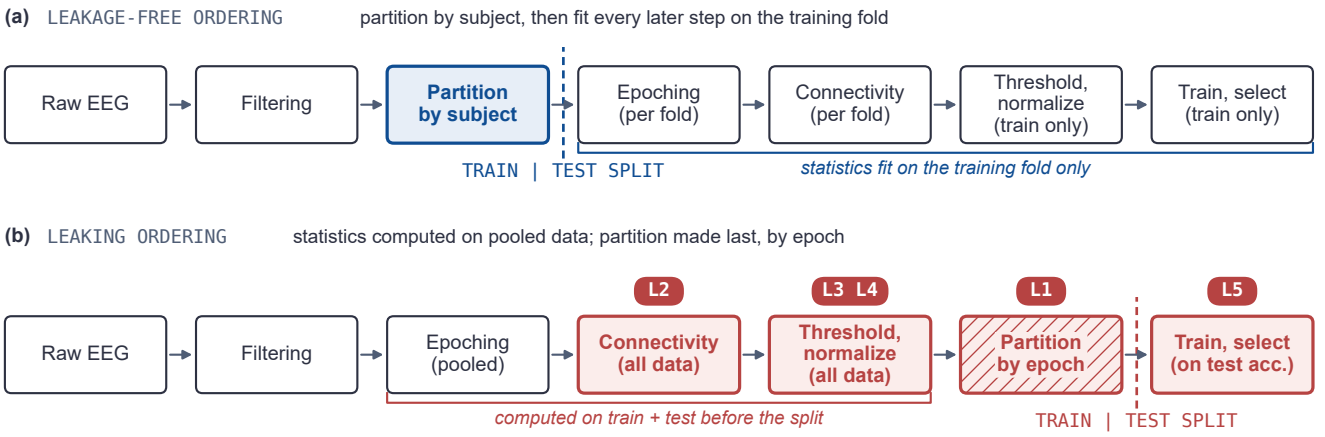

(a) LEAKAGE-FREE ORDERING
partition by subject, then fit every later step on the training fold
Raw EEG
Filtering
Partition by subject
Epoching (per fold)
Connectivity (per fold)
Threshold, normalize (train only)
Train, select (train only)
TRAIN | TEST SPLIT
statistics fit on the training fold only
(b) LEAKING ORDERING
statistics computed on pooled data; partition made last, by epoch
Raw EEG
Filtering
Epoching (pooled)
L2
Connectivity (all data)
L3 L4
Threshold, normalize (all data)
L1
Partition by epoch
L5
Train, select (on test acc.)
computed on train + test before the split
TRAIN | TEST SPLIT
L1 epoch-level split
L2 connectivity scope
L3 threshold selection
L4 normalization
L5 channel or band selection